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Risk assessment in autonomous driving: a comprehensive survey of risk sources, methodologies, and system architectures

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Abstract

As autonomous driving technology advances from assisted to higher levels of autonomy, the complexity of operational environments and the uncertainty of driving tasks continue to increase, posing significant challenges to system safety. The key to ensuring safety lies in conducting comprehensive and rational risk assessments to identify potential hazards and inform policy optimization. Consequently, risk assessment has emerged as a critical component for ensuring the safe operation of higher-level autonomous driving systems. This review focuses on research into risk assessment for autonomous driving. It systematically surveys the state-of-the-art literature from three key perspectives: risk sources, assessment methodologies, data foundations, and system architectures. For each perspective, the paper provides an in-depth analysis of representative technical approaches, modeling principles, and typical application scenarios, while summarizing their research characteristics and applicable boundaries. Finally, this paper synthesizes the three fundamental challenges that persist in current research and further explores future directions and development opportunities. It provides a theoretical foundation and methodological references for the development of autonomous driving systems that exhibit high safety and reliability.

Keywords: Autonomous driving, Risk assessment, Safety evaluation

1 Introduction

Autonomous driving is an emerging interdisciplinary field that has increasingly become a central focus of global technological innovation [1, 2]. Its development is propelled by advances in automotive electronics, computer science, artificial intelligence, and related disciplines. In the context of intelligent transportation systems, autonomous driving contributes to improved traffic efficiency, significantly reduces accidents caused by human error, and fundamentally reshapes traditional modes of transportation [3]. As the technology continues to advance, its large-scale application is fostering the growth of innovative industries and exerting a profound influence on the global economy.

In recent years, autonomous driving systems have gradually advanced from human-centric Advanced Driver Assistance Systems (ADAS) to higher levels of driving automation [4]. As their operational design domains (ODDs) continue to expand, these systems face growing challenges related to environmental complexity and task uncertainty. For instance, highway driving and urban driving present markedly different demands. In urban environments in particular, dynamic agents such as pedestrians and cyclists, along with variable traffic signals and complex road structures, require autonomous systems to possess a high level of perception accuracy, decision-making robustness, and adaptability.

In this context, ensuring the safety of autonomous driving systems is becoming increasingly challenging, highlighting the critical role of risk assessment [5]. A scientifically grounded risk assessment framework can effectively identify potential hazards and anticipate safety issues in

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advance. It supports the detection of risks across system design, perception, and decision-making modules, providing valuable input for system refinement. This enables autonomous vehicles to make safe and reliable decisions in complex and dynamic environments. Therefore, comprehensive and systematic risk assessment serves as a fundamental basis for the safety assurance of autonomous driving systems [6].

Building on this foundation, this paper provides a comprehensive survey of risk assessment research in autonomous driving. It investigates the essential role of risk assessment in enhancing system safety from three key perspectives: the risk source perspective, which explores the types of uncertainty and their underlying generation mechanisms; the risk assessment methodology perspective, which examines the transition from rule-based approaches to learning-based modeling paradigms; and the data and assessment framework perspective, which focuses on the acquisition channels and representation formats of high-value risk-related data, as well as the system-level validation of assessment outcomes. The paper further summarizes three major challenges currently facing the field, including the absence of a unified risk metric system, the limited consideration of system capability in risk estimation, and the scarcity of high-quality labeled risk data.

Figure 1 illustrates the classification scheme of risk quantification methods for autonomous driving, organizing existing approaches into key dimensions. This structured framework helps clarify the landscape of current methodologies and provides a solid foundation for systematic analysis, allowing for a deeper understanding and comparison of the strengths and limitations of each approach. In the following sections, this paper will define the risk evaluation problem in more detail.

2 Problem description

The core goal of the risk assessment model is to quantify the risk level faced by the autonomous driving system through a comprehensive analysis of the vehicle's current state and environmental situation, and predict the potential threats that these risks may pose to driving safety. On this basis, the results of risk assessment can serve as an important basis for safety decision-making and behavior adjustment of the autonomous driving system, guiding it to take timely response measures, thereby effectively avoiding or reducing the occurrence of risk events.

To achieve this goal, the risk assessment model needs to fully consider the dynamic interaction factors of vehicles in complex traffic environments, accurately capture the uncertainty and temporal evolution of multi-agent behaviors, and comprehensively evaluate the impact of the external environment on system safety. The most intuitive way to assess whether the operating state of an autonomous driving system is safe is to analyze the impact of the surrounding traffic environment on its behavioral capabilities, especially the changing trends of the states and behaviors of surrounding traffic participants.

However, it is worth noting that the risk level is not only determined by the external environment but also closely related to the autonomous driving system's own response capabilities. In the same traffic scenario, different driving subjects may face completely different risk levels due to differences in capabilities such as perception accuracy, decision-making response speed, and control robustness. Experienced drivers or superior autonomous driving algorithms can usually more effectively identify potential risks and take reasonable responses, thereby maintaining a high level of safety; while systems with weaker capabilities may be exposed to higher risks due to delayed responses or strategic mistakes.

Therefore, in order to more comprehensively and accurately assess the safety of autonomous driving systems, the risk assessment model should incorporate algorithm capabilities as a key variable into the modeling paradigm, and realize risk modeling and assessment oriented to capability differences from the two aspects of environmental complexity and system capabilities. In summary, the risk assessment model can be defined as:

$$\Lambda = \left\{ f_{RQ}\left(s_{RQ}, A_{\theta_\tau}\right) \right\}, \tag{1}$$

where s_{RQ} represents the state space of the risk assessment model, which is used to characterize the surrounding traffic environment situation. A_{θ_τ} denotes the driving model, where θ_τ stands for the policy parameters. This formula indicates that the current state input s_{RQ} and the driving model A_{θ_τ} jointly influence the result Λ of risk assessment.

As shown in Fig. 2, risk assessment plays a crucial role in the autonomous driving system, interacting with the perception-decision-control pipeline. Figure 2 illustrates how risk assessment is integrated into each stage of the system, influencing decision-making and control strategies to ensure safe and efficient driving.

In the perception phase, risk assessment evaluates the information received from sensors (such as cameras, LiDAR, and radar) to assess the environment's safety. During the planning phase, the system uses risk-based metrics to evaluate potential actions and select the safest option, considering factors such as collision risk and road conditions. Finally, in the control phase, the system's decision is translated into actual control commands, with real-time risk monitoring ensuring that the vehicle responds dynamically to any changes in its environment.

The above problem definition lays a unified analytical foundation for the subsequent review. Table 1 provides a structured summary of representative risk quantification methods, categorized from four key perspectives: risk

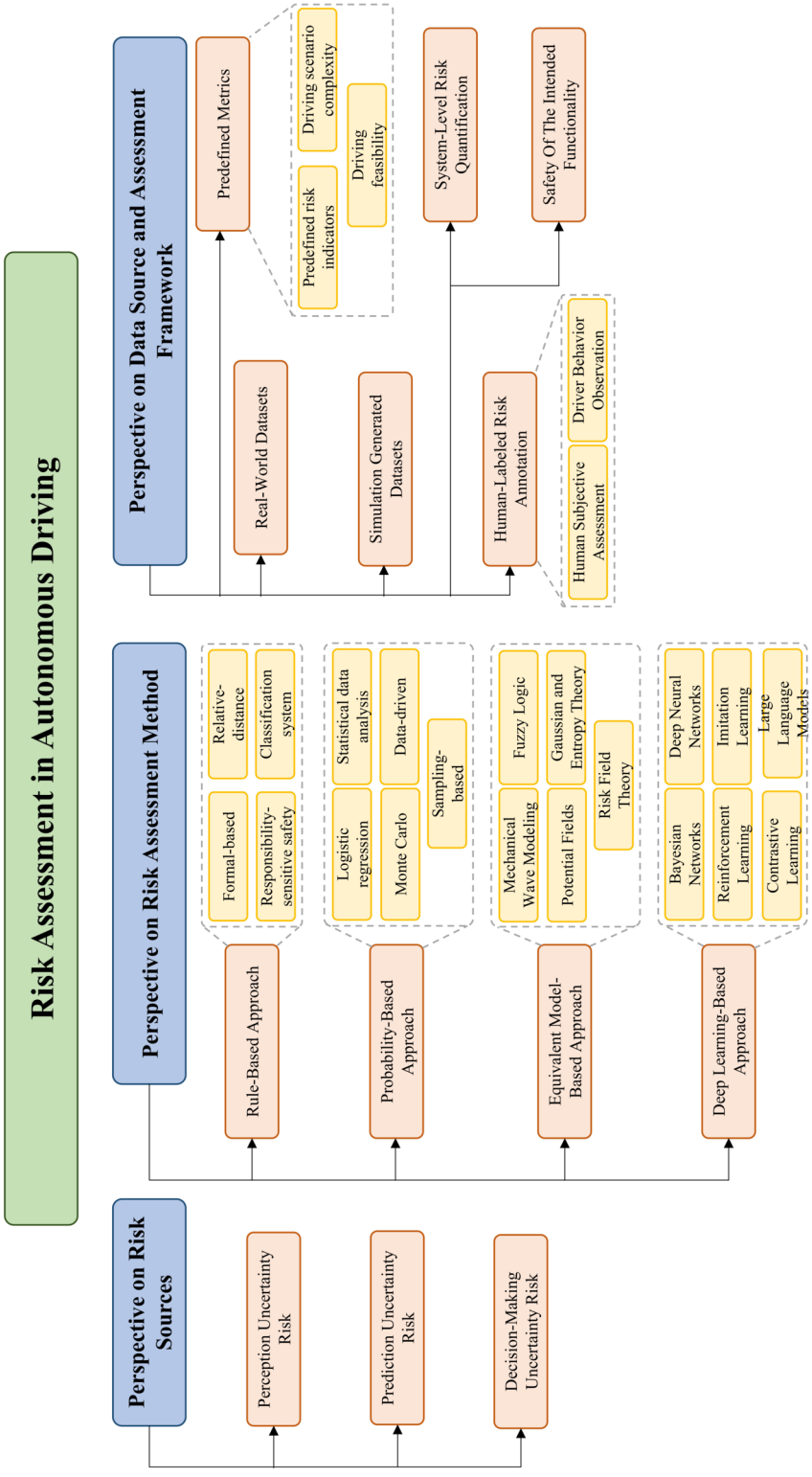


Figure 1 Classification scheme of risk quantification methods for autonomous driving. The classification encompasses three key dimensions: risk sources, risk assessment methodologies, data sources and overarching assessment frameworks

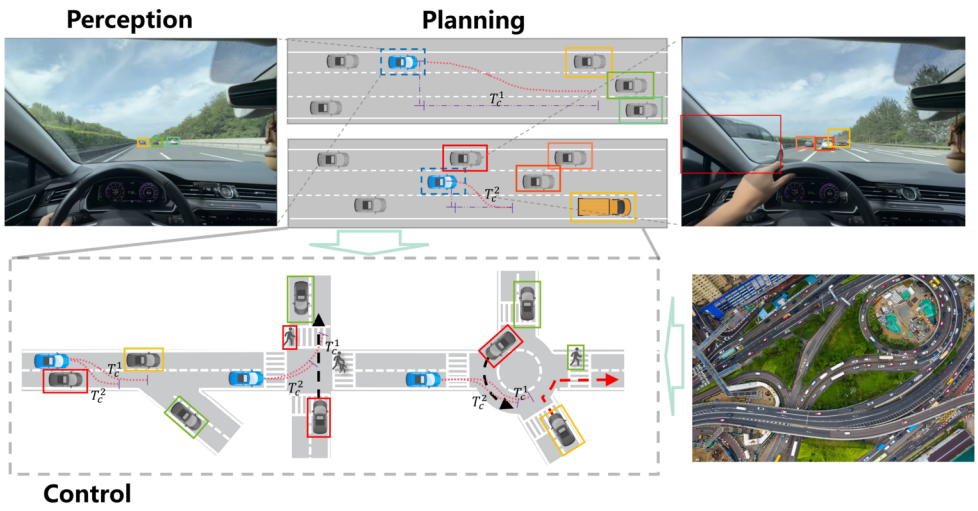


Figure 2 The integration of risk assessment within the perception-planning-control pipeline of an autonomous driving system

sources, assessment methodologies, data sources, and assessment frameworks. This classification serves as a reference for understanding the modeling paradigms and technical focuses of existing approaches.

3 Perspective on risk sources

In the context of autonomous driving systems, safety concerns emerge from failures or deviations in critical components, including perception, prediction, and decision-making processes. Significant information dependencies and structural coupling between modules within the system facilitate the transmission and amplification of risks through the coupled structure between modules. Therefore, classifying risks from the perspective of their sources helps systematically identify the root causes of uncertainty and provides a theoretical basis for designing subsequent risk mitigation strategies.

Specifically, from this perspective, risks can be categorized into three types: perception uncertainty risks, primarily caused by the limitations of sensor performance and information ambiguity or loss due to complex environmental interference; prediction uncertainty risks, manifested as difficulties in inferring the future behavior of traffic participants and insufficient coverage of long-tail scenarios; and decision uncertainty risks, focusing on the multiple conflicts and uncertainties faced when generating safe and feasible control strategies under incomplete information. These three categories of risks constitute the critical weak points in the operation of autonomous driving systems, and their systematic assessment is a prerequisite for establishing a reliable risk control system.

3.1 Risk induced by perceptual uncertainty

The real world is characterized by complexity and variability, and sensors (including cameras, lidar, and millimeter-wave radar) are inevitably subject to physical constraints and environmental interferences. The resulting risk of perceptual uncertainty refers to the ambiguity, incompleteness, or even errors inherent in the system's comprehension of the environment. For example, it may be unable to accurately recognize partially occluded pedestrians, misclassify or miscalculate the distance of distant objects, or lose lane line information under rainy or foggy conditions. Such uncertainty, originating from the acquisition and interpretation of raw data, constitutes the most fundamental and fragile link in the safety chain of autonomous driving. Its errors can directly propagate and amplify to subsequent prediction and decision-making modules, thereby forming a fundamental safety hazard. The following section will introduce research methods for quantifying such uncertainty.

In autonomous driving systems, the impact of perceptual uncertainty on safety has emerged as a critical issue in research. To quantify the influence of perceptual uncertainty on the safety of autonomous driving, the reference [43] constructs a sensor uncertainty fault tree and proposed a comprehensive safety risk assessment method based on probability models. By constructing a fault tree analysis of sensor detection uncertainty, combined with target distance distribution, collision probability, and passenger injury risk models, the impact of perceived uncertainty on autonomous driving safety was systematically quantified, providing multidimensional quantitative basis for safety assessment. In connection with this, the reference [44] proposes a model that integrates sequence inference and virtual agents. By dynamically tracking occluded

Table 1 Classification of risk quantification methods for autonomous driving

Perspective	Method Categories	Representative Methods	Description
Risk Sources	Perceptual Uncertainty	PDQ, AHRL, PURE, EMUFormer, TUQCP [7–11]	Quantifies risk stemming from inaccuracies or noise in the perception module
	Prediction Uncertainty	EIGNN-GD, IAIMM-KF, CATPlan [12–14]	Captures uncertainty in predicting future behavior of surrounding agents
	Decision-Making Uncertainty	EQN, MABAC [15, 16]	Assesses risk due to variability or ambiguity in decision outputs
Assessment Methodology	Rule-Based Methods	iRisk, SafeCast, SRD [17–19]	Rely on expert knowledge or pre-defined rules to evaluate risk
	Statistical and Probabilistic Methods	EVORA, PURE, DRS-RL [9, 20, 21]	Use probabilistic models or statistical analysis, such as logistic regression
	Equivalent Model-Based Methods	SF-ANP, LightGBM, Double-M, RiskNet [22–25]	Model risk by constructing abstract representations such as potential fields or mechanical analogies
	Learning-Based Methods	DBNID, SeNeVA, RACER [26–28]	Leverage deep learning or reinforcement learning models to estimate risk
Data Sources	Large-Scale Dataset Filtering	CODA, RBFVDS [29, 30]	Identify safety-critical scenarios from massive real-world driving datasets
	Simulation-Based Scenario Construction	DeepMF, CBN [31, 32]	Generate rare or high-risk scenarios through simulation for risk evaluation
	Human-Driven Data Collection	DEA, RES-SE-CNN [33, 34]	Derive risk-related data through human behavioral analysis
Assessment Framework	Indicator-Based Evaluation	CALMM-Drive, DSCQ, AHP-TOPSIS [35–37]	Construct frameworks based on pre-defined evaluation indicators
	System-Level Quantification	OOD [38, 39]	Assess risk by quantifying the overall operational performance of the system
	Safety of the Intended Functionality (SOTIF)	SOTIF Entropy, Multi-Dimensional Model, The SOTIF Meta-Algorithm [40–42]	Establish frameworks in accordance with ISO 21448 to address functional insufficiencies under nominal conditions

areas, generating multiple types of virtual agents, and predicting their future behavior, it achieves more accurate risk assessment and decision optimization while avoiding overly conservative decisions.

With the continuous introduction of new technologies and methods, the accuracy and efficiency of perceptual uncertainty quantification have been significantly improved. By integrating techniques such as deep learning, Bayesian inference, and Monte Carlo methods, researchers have proposed a variety of effective quantification approaches to enhance the capability of autonomous driving systems in perceiving uncertainty and risks. Reference [45] introduces deep learning anomaly detection into the vehicle subsystem, quantifies perceptual uncertainty based on sensor data association, and provides support for predictive maintenance and robustness evaluation; In the field of object detection, the reference [7] embeds the Monte Carlo Dropout technique into the You Only Look Once version 5 (YOLOv5) model, models the output fluctuations through Gaussian parameters, achieves joint uncertainty quantification of target category and localization, and ensures real-time performance; The Prediction sURface uncErtainty (PURE) method proposed in the reference [9] further extends this technique by improving the YOLO and Single Shot MultiBox Detector (SSD) series models to adapt to Monte Carlo Dropout, and combining Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering prediction center and convex hull area calculation to quantify the uncertainty of the predicted surface. Based on the evidence theory, the reference [46] proposes the EviSeg method, which innovatively reconstructs the pixel classification output of the fully convolutional network as an evidence pool, calculates conflict metrics using convolution parameters, and significantly improves computational efficiency while ensuring performance.

In the field of 3D perception, the reference [47] proposes a framework that integrates uncertainty quantification and propagation, constructs a Bayesian Point-based 3D Object Detection from Point Clouds (PointRCNN) probabilistic 3D object detector, quantifies cognitive uncertainty through Monte Carlo Dropout, directly models and quantifies random uncertainty, converts it into a spatial ellipse, and combines it with an opportunity constrained motion planner to evaluate collision risk. In terms of efficient quantification schemes, the reference [10] proposes the EMUFormer teacher-student distillation method, which utilizes the prediction uncertainty of the teacher model to achieve efficient multi task uncertainty quantification in the joint task of semantic segmentation and monocular depth estimation, with performance comparable to deep integration methods.

To comprehensively enhance the risk perception and decision-making capabilities of autonomous driving systems in complex environments, current research not only

focuses on quantification methods for perceptual uncertainty but also gradually expands into such directions as the analysis of uncertainty sources, the enhancement of perceptual robustness in adversarial scenarios, and the risk modeling and reasoning for occluded regions. Reference [48] systematically analyzes the sources of uncertainty in autonomous driving perception (including distortion caused by internal and external factors of sensors and complex scenes), divides them into two categories: cognitive and random, sorts out four mainstream quantitative methods (single deterministic method, Bayesian method, ensemble method, and test enhancement), and proposes evaluation criteria that cover accuracy and efficiency. For adversarial attack scenarios, the Trusted Uncertainty Quantification in Collaborative Perception framework (TUQCP) proposed in the reference [11] perturbs intelligent agents to share information through adversarial training, combines learning quantization modules with normal prediction calibration, introduces uncertainty loss terms to optimize detection accuracy, and forms a collaborative detection perception uncertainty quantization scheme.

In the modeling of occlusion risk, the Adaptive Heuristic Reinforcement Learning (AHRL) method proposed in the reference [8] improves Dueling Double Deep Q Network (D3QN) to obtain the Dueling Double Deep Q Network (Adapt-HD3QN) algorithm, incorporates heuristic functions to improve efficiency, and uses a mixed artificial potential field to handle obstacle avoidance. The collision of occlusion areas is modeled as a zero sum game, and risk quantification is achieved using dynamic Bayesian networks. Finally, the reference [49] proposes an ontology based approach to quantify the risk of perception defects in autonomous driving by constructing trigger sources and perception stage ontologies, defining conceptual relationships, and generating trigger conditions based on event propagation chains, combined with worst-case assumptions and probabilistic risk assessment.

3.2 Risk induced by prediction uncertainty

The risk of predictive uncertainty stems from the inherent challenges in inferring future states. The behaviors of traffic participants are characterized by high randomness, interactivity, and unknowable intentions. Therefore, the prediction of autonomous vehicles not only relies on the currently observed states but also needs to understand potential intentions and consider complex interactions among multiple agents. Furthermore, the limitations of prediction models themselves, their dependence on the distribution of historical data, and insufficient coverage of rare or long-tail scenarios all affect prediction results. Consequently, how to effectively model, quantify, and manage predictive uncertainty, and formulate safe decision-making strategies based on this, has become particularly crucial. The following will introduce methods for quantifying and evaluating the risk of predictive uncertainty.

In terms of dynamic interaction modeling, the reference [12] combines entropy aware graph neural networks with Gaussian distributions to achieve dynamic modeling of vehicle spatiotemporal interactions and quantification of behavioral uncertainty; Another reference [50] proposes a multi view hypergraph contrastive learning framework that integrates deterministic deep learning with probabilistic assumptions to optimize risk assessment. In response to complex interactions in dynamic scenarios, the reference [51] combines aggressive driving models with incomplete information game theory to dynamically estimate collaborative risks through trajectory prediction and environmental assessment; Meanwhile, in the field of Model Predictive Control (MPC), the reference [52] introduces virtual obstacles and spatiotemporal projection probability risk distribution to quantify the dynamic uncertainty of occluded areas in autonomous driving.

Regarding the issue of decomposing the uncertainty inherent in prediction results, the reference [53] uses a deep learning model based on two-dimensional histograms combined with deep ensemble techniques to measure arbitrariness and cognitive uncertainty through entropy based quantitative indicators. Reference [54] adopts a specific framework to separate data and model uncertainty using context cross embedding and dual channel learning, and dynamically quantifies traffic prediction risks through spatiotemporal propagation networks. Reference [55] utilizes deep learning prediction models and spatial probability estimation to quantify the nonlinear evolution of uncertainty in autonomous driving prediction.

With the deep integration of prediction modules and control & planning modules, uncertainty modeling is gradually extending from the level of individual tasks to the level of system-wide collaboration. Reference [13] utilizes interactive perception multi model prediction and online sampling techniques to embed obstacle motion uncertainty into MPC through Gaussian mixture models, achieving a balance between safety and performance. Regarding the core issues in the prediction process, the reference [56] proposes an uncertainty scoring framework that integrates motion and model uncertainty, and quantifies prediction uncertainty through ensemble learning and evidence learning; There is also the reference [57] to construct a benchmark dataset for multi-source uncertainty in autonomous driving (MUAD), which systematically quantifies the dual uncertainty of data and models using methods such as deep integration and Monte Carlo dropout. In addition, the reference [14] proposes a lightweight loss prediction module CATPlan, which decodes motion and planning embedding as collision loss and quantifies trajectory uncertainty as binary collision risk, further improving the quantification system of prediction risk.

3.3 Risk induced by decision uncertainty

Decision uncertainty risk refers to the risk that arises when generating real-time control commands that are safe, comfortable, and compliant with traffic rules, wherein it is difficult to determine the optimal or safe decision itself. This difficulty stems from factors including: the incomplete observability of the environment state, uncertainty in action execution, potential conflicts among task objectives, and the challenge of precisely evaluating long-term consequences arising from extended decision sequences.

To address the risk assessment challenges posed by multi-source heterogeneous data and complex decision-making environments, the integration of multi-criteria decision-making (MCDM) methods with multi-faceted uncertainty modeling is emerging as a promising research direction to enhance risk quantification capabilities at the decision-making level of autonomous driving. Reference [16] constructs a hybrid MCDM framework that integrates the Analytic Hierarchy Process (AHP), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), and fuzzy sets to quantify the physical and network attack risks under the six major standards of autonomous driving; At the same time, modeling and integrating uncertainty in the process of risk quantification has become a key breakthrough point: Reference [58] models random uncertainty through deep reinforcement learning (DRL) and integrates relevant models to quantify cognitive uncertainty, thereby establishing an adjustable threshold framework for intersection autonomous driving decision risk; In addition, the reference [59] focuses on the sources of uncertainty in perception, prediction, and decision-making systems, directly quantifying the decision-making risks of autonomous driving; The Ensemble Quantile Networks framework proposed in the reference [15] improves the accuracy of risk quantification by separating two types of uncertainties into quantitative decision-making.

Reference [60] uses Polynomial Chaos Expansion to quantify the uncertainty of target location in the perception phase, and integrates its propagation effect into the decision-making process through dynamic Bayesian networks and influence diagram models, effectively enhancing the safety and interpretability of the auto drive system. For dynamic decision-making scenarios, the reference [61] evaluates vehicle interaction behavior by constructing an asymmetric driving aggression model, analyzes lane changing decisions using game theory, and uses stochastic model predictive control (MPC) to address uncertainties such as communication delay and motion variability, achieving the quantification of dynamic risks in autonomous driving decision-making.

4 Perspective on risk assessment method

Building on the systematic identification and categorization of risk sources, achieving reasonable risk assessment has become a key challenge in current research on autonomous driving safety. To comprehensively understand the modeling mechanisms and applicable boundaries of different technical approaches, it is essential to systematically analyze and classify existing studies from the perspective of specific technical methods.

This section divides risk assessment methods into four categories: first, rule-based methods, which emphasize formal constraints and verifiability, and are suitable for structured scenarios with clear rules; second, probability-statistics-based methods, which model the probability distribution of risks using historical data, focusing on the probabilistic characterization of uncertain events; third, equivalent-model-based methods, which establish risk propagation relationships in temporal and spatial dimensions through physical or mathematical abstractions; and fourth, deep-learning-based methods, which rely on large-scale data and neural network modeling capabilities to integrate risk representation and decision-making processes.

It can be seen that risk assessment methods are undergoing an evolution from rule-driven to data-driven approaches, and various methods exhibit different technical trade-offs and development trends in terms of generalization ability and adaptability to application scenarios, as summarized in Table 2.

As shown in Fig. 3, it presents a visual comparison of different risk assessment methods. Figure 3(a) is the third-person view of the roundabout scene, Fig. 3(b) shows the visualization results of the artificial potential field method, and Fig. 3(c) presents the visualization results of the data-driven method. From different perspectives, the figure illustrates the risk assessment methods: Fig. 3(b) demonstrates the application of the potential field method in risk assessment, visually presenting the risk variation process of the vehicle in a complex environment through the distribution of the force field; while Fig. 3(c) shows the risk assessment results based on the data-driven method, highlighting the richness of information and adaptability of data-driven methods when handling complex scenarios.

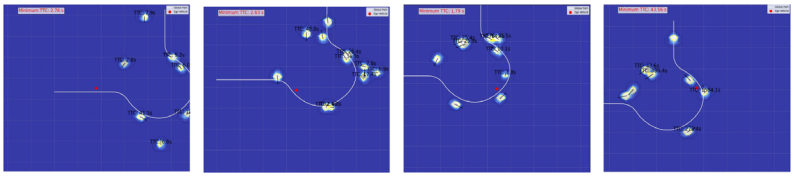
4.1 Rule based methods

Risk assessment methods based on preset rules establish explicit prior constraints to define boundaries and perform logical reasoning for potential risks in autonomous driving scenarios. Such methods possess favorable interpretability and formal verification capabilities, and are particularly applicable to closed or structured environments with clear traffic rules and well-defined behavioral norms. Their fundamental idea lies in transforming safety requirements into binding logical or mathematical conditions—for example, characterizing behavioral sequences using temporal logic, or setting spatial safety boundaries through geometric relationships—thereby enabling the deduction

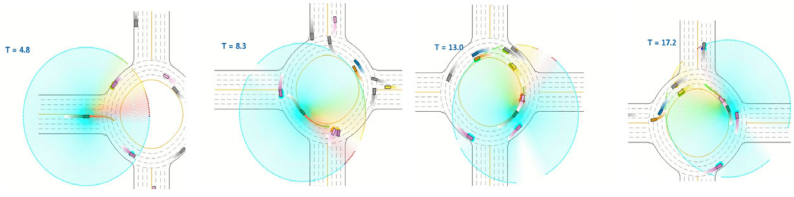
Table 2 Risk Assessment Methods and Their Application Scenarios

Risk Assessment Method	Typical Driving Scenarios
Rule-based Methods	Structured scenarios with clear rules (e.g., highway, controlled environments)
Probability-Statistics-based Methods	Uncertain environments where historical data can be used to predict risk
Equivalent-model-based Methods	Scenarios where risk propagation is influenced by physical or mathematical relationships (e.g., intersection, highway merging)
Deep-learning-based Methods	Complex, dynamic environments with large-scale data availability (e.g., urban environments)

(a) Third-person view of the roundabout scene.



(b) Visualization of the artificial potential field method.



(c) Visualization of the data-driven method.

Figure 3 Visual comparison of different risk assessment methods

and verification of the safety of system behaviors. Typical approaches include formally complete model checking mechanisms and the normative design of spatiotemporal safety intervals in the Responsibility-Sensitive Safety (RSS) model.

This section will elaborate on the representative practices of this category of methods, starting from early static assessment strategies dominated by fixed rule constraints, gradually transitioning to dynamic assessment mechanisms that can adjust rules according to scenario changes, and further exploring the practical challenges they face in complex open environments, such as insufficient generalization ability and limited adaptability of rules.

From the perspective of logical modeling and formal expression, the reference [62] extends the human like risk index and integrates it into the Linear Time Series Logic (LTL) specification, transforming the control synthesis problem into an reachability problem. It constructs a linear programming model using occupational metrics and introduces soft and hard risk thresholds to balance different risks, thereby quantifying the perceived uncertainty of autonomous driving. Regarding the core path of risk quantification, the reference [63] proposes two basic methods: one is based on collision prediction, which detects the possibility of collision by predicting the future trajectory of the vehicle, covering risk indicators such as binary collision prediction and probability collision prediction; Another type is based on abnormal behavior detection, which evaluates risks by identifying vehicle behaviors or conflicts that deviate from the norm. At the same time, innovative risk models have been developed for specific scenarios, such as the concise and effective Potential Damage Risk (PODAR) model proposed in the reference [64]. Based on the publicly available dataset of obstacle avoidance experiments, an individual risk perception model has been established for highly adaptive drivers, revealing that differences in risk perception stem from drivers' assessment of potential damage under spatiotemporal uncertainty.

To enable interpretable modeling of safety boundaries in autonomous driving systems, the Responsibility Sensitive Safety (RSS) theory has been widely adopted. Proposed by Mobileye, the RSS model is designed to delineate the boundaries of safe driving behaviors through formalized rules, thereby ensuring the mitigation of ambiguities in responsibility attribution within multi-agent interaction scenarios. As a theoretical framework dedicated to rule-driven safety modeling, RSS establishes a verifiable foundation for behavioral constraints in autonomous driving systems by modeling safety distances and response protocols for both longitudinal and lateral behaviors. In recent years, researchers have further extended the application scenarios and risk assessment capabilities of RSS, developing various integrated models to support safety evaluation in complex environments.

Reference [65] proposes a general objective quantification framework to describe the spatiotemporal interaction between vehicles from three dimensions: quantity, diversity, and relationship. This framework is based on the Responsibility Sensitive Security (RSS) theory and identifies surrounding vehicles that contribute to environmental complexity through dynamic impact area recognition. This framework quantifies human drivers' judgments on the complexity of the driving environment, thereby achieving risk quantification in autonomous driving. Among them, the reference [66] extends the RSS model to construct a comprehensive risk index, integrates vertical and horizontal risk indicators, uses segmented linear functions to measure the possibility of vehicle collisions, and incorporates risk preference parameters to reflect different risk attitudes. After verification by the Unity engine, this index is positively correlated with simulated collision events and can support training and validation in autonomous driving scenarios. Reference [19] further combines RSS theory with decision models: SRD motion decision model integrates RSS safety module and lane changing model through multi-objective reward function, improves traffic efficiency while ensuring safety, and finally achieves safe autonomous driving through deep reinforcement learning; Reference [18] proposes the SafeCast risk responsive motion prediction model, which integrates RSS framework encoding safety rules, introduces graph uncertainty features (GUF) to inject learnable noise into the graph attention network to capture real-world uncertainty, and generates prediction trajectories that comply with safety constraints through a multimodal decoder to complete risk quantification.

Meanwhile, some studies have approached the issue from the perspective of optimizing the testing phase, focusing on improving the generalization ability of models prior to deployment. Reference [67] proposes the Test Time Augmentation (TTA) method, which generates multi version predictions by enhancing the color of images. Combined with independent optimization of bounding box selection and clustering criteria for classification and localization subtasks, it improves the accuracy of pre trained target detectors while achieving uncertainty quantification. For the car following scenario, the reference [68] conducts real car experiments using a fixed base driving simulator. The scale estimation method was used to explore the subjective risk perception of drivers towards the variables of headway and collision time. The physical parameters were correlated with psychological risk indicators, and the risk perception was quantified as the sum expression of the reciprocal of headway and the reciprocal of collision time. The effectiveness was verified by real car data.

Similarly, several studies have developed targeted and interpretable risk assessment models by incorporating environmental uncertainty factors, driving behavior characteristics, and rule constraint mechanisms. These methods are no longer confined to modeling single risk factors under static conditions; instead, they emphasize the systematic characterization of risk generation mechanisms and evolutionary processes based on multi-source information in real-world road scenarios. Reference [69] proposes a formal security method based on deep reinforcement learning. By constructing an environmental model that includes the influence of non deterministic environmental factors to quantify risk, it is used to formalize the reward machine structure to guide the setting of reinforcement learning reward functions and generate control obstacle functions to ensure policy security; In addition, the reference [70] proposes the critical closed-loop framework, which integrates traffic dynamics, driving behavior analysis, alternative safety measures, and optional large language model components. Through closed-loop feedback of data generation and training processes, it quantifies risks and accurately generates key scenarios for performance gaps in reinforcement learning agents. Reference [71] defines the lane change safety factor and constructs a risk assessment model based on the hyperbolic tangent function by combining the characteristics of the coefficient changes. Environmental adjustment factors are introduced to consider the impact of weather and road conditions, and the analytic hierarchy process is used to determine the weights of the influencing factors.

4.2 Probability and statistics methods

This section reviews the application of probabilistic-statistical methods in risk modeling for autonomous driving. Such methods treat potential risks as random events and achieve quantitative assessment of risk levels by analyzing historical data or simulation samples to infer the probability distribution of conflict occurrences. Compared with rule-driven approaches, probabilistic models

are more suitable for scenarios dominated by uncertainty, and exhibit stronger modeling capabilities especially when dealing with implicit conflicts or long-tail risks. This section will systematically sort out the progress of representative risk quantification technologies from the perspective of two mainstream method categories.

Firstly, some studies have proposed various methods for capturing the uncertainty structure of systems and conducting risk assessments based on statistical modeling approaches. These methods are typically built upon explicit probabilistic assumptions or distribution modeling frameworks. They transform potential risk factors in the process of autonomous driving into quantifiable random variables, and depict the risk distribution characteristics of the system under different states through statistical inference on observed data. In research based on logistic regression models, the Prediction Surface Uncertainty (PURE) method was proposed by the reference [9], which improves the object detection model to generate multi bounding box predictions, combines Density-Based Spatial Clustering of Applications with Noise (DBSCAN) clustering and convex hull area calculation to quantify uncertainty, and provides a basis for safety decision-making; The cloud model security assessment method proposed in the reference [72] integrates Time to Collision (TTC) and danger threhold (DT) indicators to generate a risk cloud map, and determines the risk level through proximity analysis. Reference [20] proposes the evidential off-road autonomy (EVORA) framework based on statistical data analysis, which utilizes evidence learning and normalized flow networks to capture two types of uncertainties, and combines Conditional Value-at-Risk(CVaR) to achieve safe navigation in off-road scenarios; Reference [73] quantifies lane keeping uncertainty through MC dropout and uses genetic algorithm to generate test cases to evaluate model robustness; Reference [74] is based on data-driven methods to construct a Model Predictive Control (MPC) framework for risk quantification. This framework innovatively combines Kernel Principal Component Analysis (KPCA) for nonlinear dimensionality reduction to capture key system features, uses Kernel Density Estimation (KDE) for non parametric state transition probability estimation, and uses Hankel matrices to characterize the dynamic temporal correlation of the system.

Building upon the aforementioned methods, sampling - enhancement techniques have further improved the generalization ability and expressiveness of risk quantification systems. Traditional statistical models often rely on fixed data distributions or simplified assumptions. When faced with complex and dynamic environments, they may fail to fully capture long - tailed risks and occasional conflicts. In contrast, sampling - based methods introduce a large number of perturbed samples, simulated trajectories, or environmental changes, enabling the system to cover more potential risk scenarios. This effectively compensates for the limitations caused by insufficient training data or skewed distributions. The sampling based technology has further improved the risk quantification system. Reference [75] proposes the Convolutional Conditional Neural Processes (Conv NPs) method, which optimizes environmental perception through point cloud upsampling and uncertainty filtering, supporting safe navigation and map construction; The Dynamic Residual Safe Reinforcement Learning (DRS-RL) framework in the reference [21] innovatively proposes the multi-agent dynamic conflict zone model to quantify the spatiotemporal coupling risk of heterogeneous traffic participants, combining weak to strong safety correction paradigm and risk perception replay mechanism to dynamically calibrate safety boundaries and reduce data bias, achieving accurate quantification of risks in autonomous driving decision-making. These methods focus on multidimensional aspects such as model improvement, data analysis, and algorithm optimization, providing comprehensive technical support for the safety risk assessment of autonomous driving.

In the research on risk quantification for autonomous driving, uncertainty modeling has gradually emerged as a key direction for enhancing the safety and robustness of systems. In particular, studies that integrate tools such as probability distributions and information entropy have demonstrated excellent capabilities in risk perception and expression. Reference [12] proposes an EIGNN-GD framework based on entropy aware graph neural network (EIGNN) and Gaussian distribution (GD). The Gaussian distribution is used to probabilistically model vehicle speed and acceleration, entropy theory is introduced to quantify risk uncertainty, and graph neural network is used to capture the spatiotemporal dynamic characteristics of multi vehicle interaction, achieving accurate quantification of collision risk in complex traffic scenarios and providing technical support for autonomous driving safety decision-making. Another reference [24] proposes the "Double-M Quantification" method, which directly models the bounding box corners through customized mobile block bootstrap algorithm and multivariate Gaussian distribution. In a single inference process, it captures both cognitive uncertainty and random uncertainty in collaborative target detection, completing the risk quantification of autonomous driving and providing support for subsequent planning and control modules to achieve safe driving.

To further model scenario uncertainty and perception uncertainty in autonomous driving, some studies have introduced fuzzy theory to enhance the ability to express complex environments and ambiguous information. Regarding the quantification of scene and perceptual uncertainty, some studies have further expanded the application dimensions of fuzzy theory. Among them, the

spherical fuzzy network analysis method (SF-ANP) reference [22] risk assessment method quantifies scene uncertainty (risk) through a five layer model covering the self vehicle, road, traffic infrastructure, surrounding objects, and environment, using fuzzy sets and element interaction weights, providing a basis for automatic driving test scene classification and safety function verification. Reference [22] proposes a dual layer object detection method inspired by human perception, which focuses on improving perceptual reliability. The first layer scans the road, and the second layer identifies objects in key areas; By integrating the fuzzy logic system with the key area recognizer for path planning, the areas that require secondary detection are determined, which improves accuracy and optimizes the allocation of computing resources while reducing perceived uncertainty risk, ensuring the safety of autonomous driving. Reference [76] proposes an Autonomous Fuzzy Decision Support System (AFDSS), which reduces data complexity through self-organizing maps (SOM), extracts risk factors from massive unlabeled data, evaluates risk levels using a fuzzy inference system, predicts risks based on the similarity between new and existing events, and conducts sensitivity analysis through optimistic, pessimistic, and neutral strategies to provide support for safe driving decisions. Another reference [77] builds a risk framework based on fuzzy logic, combines attack feasibility and security impact, builds a reasoning system containing fuzzy sets and natural language rule base, quantifies input uncertainty into continuous risk values, and provides a more detailed basis for network security risk assessment of auto drive system.

4.3 Equivalent model method

In the equivalent model approach, complex traffic systems are simplified into computable dynamic models through physical or mathematical abstractions (such as potential fields, wave propagation, etc.), and risk intensity is mapped in the form of continuous or discrete field domains. This type of method excels at characterizing the spatial propagation characteristics of risks. It often converts the interaction relationships between traffic participants into the changing processes of virtual force fields or energy gradients by means of mechanical wave models, risk field theory, and other techniques, thereby achieving structured modeling of potential risks.

As a representative method within equivalent modeling approaches, the Artificial Potential Field (APF) method is a classic risk assessment algorithm that is widely applied in path planning and obstacle avoidance tasks in autonomous driving [78]. This method defines obstacles and goals as potential fields with attractive or repulsive forces, simulating the interaction of different force fields to guide path selection. In this way, the system can automatically generate a safe and optimal path, with advantages such as simple computation and ease of implementation, making it particularly suitable for real-time decision-making in dynamic and complex environments.

However, the Artificial Potential Field method faces the issue of local minima, which is particularly evident in complex traffic environments. The local minimum problem refers to situations where the system may be misled during path planning, causing it to become stuck in unreachable or undesirable areas, thus leading to path stagnation or failure. This presents a challenge to the reliability and safety of autonomous driving systems. To address this issue, researchers have proposed an Improved Artificial Potential Field (IAPF) method [79], which uses various optimization techniques to effectively enhance the robustness and adaptability of the traditional method.

One common improvement is the introduction of a global navigation strategy, enabling the robot to escape local minima and continue moving toward its goal. For example, a hybrid approach combining global path planning with local obstacle avoidance can effectively resolve the issue of relying solely on local information, reducing the risk of the path falling into local minima. In addition, time-dependent artificial potential fields incorporate dynamic environmental factors (such as other vehicles, pedestrians, etc.) into the potential field model, allowing the path planning to not only reflect current static obstacles but also consider the motion states of traffic participants. This significantly improves the real-time performance of path planning and enhances the ability to handle complex dynamic environments.

Speciafically, the reference [80] builds the impact model of traffic environment elements on the auto drive system based on the potential field theory, quantifies the equivalent virtual electric quantity of each environment element through the analytic hierarchy process (AHP), calculates and weights the complexity of static elements (such as traffic signs, plants) and dynamic elements (such as pedestrians, vehicles) respectively, and realizes the quantitative assessment of traffic environment complexity, and then quantifies the uncertainty (risk) in automatic driving, providing theoretical support for safety assessment. For the risk quantification of path planning scenarios, the reference [81] proposes a path planning method based on risk assessment. By analyzing potential collision points to determine the safe lane changing range, a fifth order Bezier curve is used to generate path clusters, and an improved artificial potential field method is combined to incorporate driving style and reaction time into the path risk assessment. The optimal path is selected by comprehensively considering lane changing efficiency and comfort, achieving risk quantification of safe obstacle avoidance.

For car following scenarios, the reference [23] proposes a car following risk quantification method that considers driving style tendencies. By analogy with intermolecular

interactions, a car following model based on safety potential fields is constructed, integrating lane lines, road boundaries, and vehicle interaction potential fields; By integrating real-time risk exposure levels and risk severity, a risk quantification model is established. The fuzzy C-means algorithm is used to divide risks into four levels, and the LightGBM algorithm is used to achieve real-time prediction of car following risks, completing the quantification of uncertainty (risk) in autonomous driving. Another reference [82] is based on the molecular Morse potential field model, which analogizes the interaction behavior between vehicles to intermolecular interactions. Combined with vehicle dynamics theory, a risk quantification model is constructed considering safety margins and driving style specificity; By using a multi-layer encoder decoder long short-term memory network to predict vehicle trajectories, continuous quantification of risks for networked autonomous vehicles in scenarios such as following and changing lanes is achieved, providing a basis for safety decision-making.

Reference [83] proposes a risk prediction autonomous driving strategy based on hierarchical deep reinforcement learning, integrates risk field theory to construct an expected driving risk (ADR) index considering the weight of surrounding vehicles, designs a mixed action space containing discrete and continuous actions, combines the attention mechanism based hierarchical proximal policy optimization (AT-HPPO) algorithm, and introduces the potential collision energy in conflict (PCEC) to evaluate potential accident consequences, achieving risk quantification to support safe and efficient decision-making. The RiskNet framework proposed in the reference [25] integrates deterministic risk modeling based on field theory with probabilistic trajectory prediction based on graph neural networks (GNN). It captures multi-agent dynamics by constructing interaction fields and force propagation mechanisms, incorporates Doppler effect to achieve directional risk distribution, generates multimodal trajectory distribution to quantify behavioral uncertainty, and completes comprehensive quantification of autonomous driving risks in long tail scenarios.

Regarding the risk assessment of driving behavior and path planning, the reference [84] proposes a two-dimensional driving behavior model based on risk field. The two-dimensional scene risk is quantified within the preview time, and a third-order polynomial is used to generate a path curve that conforms to vehicle dynamics. Combined with the maximum acceptable risk, frequent adjustment of the path is reduced to achieve unified modeling of car following and lane changing behavior. The adaptive field planning framework proposed in the reference [85] constructs a dynamic risk field by improving the artificial potential field, quantifies interactive risks based on vehicle position, speed, and angle, and dynamically adjusts lane changing intentions. It simplifies curve road trajectory planning using Frenet coordinates and optimizes the trajectory generated by a fifth degree polynomial using an improved particle swarm optimization (IPSO) algorithm, achieving risk quantification while balancing safety and comfort.

In addition, the instantaneous potential energy increment (IIP) risk assessment index proposed in the reference [86] is based on the construction of a vector type safety potential field model, quantifying the relative motion risk between vehicles, and combining the aggregation method of lateral attenuation factor and positive and negative risk differentiation to enhance sensitivity to lateral risk and risk evolution, providing timely basis for safety decision-making. The vehicle road collaborative perception method based on safety field (SF) proposed in the reference [87] quantifies the strength of the safety field generated by vehicles through an ellipsoid model to assess driving risks. Combined with SF weighted point cloud downsampling technology and dynamic resource allocation strategy, it introduces delay compensation error (LCE) to evaluate the impact of transmission delay. It controls communication volume and improves perception accuracy while achieving risk quantification, fully supporting autonomous driving safety decision-making.

4.4 Deep learning method

Deep learning methods directly learn risk representations and decision - making strategies through data - driven end - to - end mapping, and are suitable for scenarios that are high - dimensional, nonlinear, and where rules are difficult to explicitly define. These methods have demonstrated excellent generalization ability and scenario adaptability in risk representation and strategy modeling, and are gradually becoming the core approach in risk assessment research.

4.4.1 Bayesian neural network method

In the research on risk assessment for autonomous driving, Bayesian methods have emerged as a key technical pathway supporting system - level safety assessment due to their advantages in uncertainty modeling, inferential expression, and interpretability. In recent years, researchers have constructed multiple model frameworks with probabilistic output capabilities by combining Bayesian theory with deep neural networks, which are applied to risk measurement in autonomous driving tasks.

On the one hand, the Bayesian framework provides theoretical support for network structure design and quantitative error analysis. Reference [88] proposes a quantitative practical Bayesian framework, which achieves objective comparisons in various aspects such as network architecture, and estimates the range of quantization errors between parameters and outputs. Reference [89]: A

Probabilistic Point-based 3D Object Detection from Point Clouds (PointRCNN)model based on Bayesian deep learning principles is constructed to quantify cognitive and random uncertainties in 3D object detection, and spatially integrate them into the PU-RRT motion planning algorithm to control collision probability through chance constraints and achieve safe driving.

On the other hand, Bayesian networks are also widely used to model the risk propagation mechanisms in perception and prediction processes. Reference [26] proposes a DBNID method based on dynamic Bayesian networks (DBN) and impact maps (ID), which quantifies the uncertainty propagation effect in perceptual prediction by combining the Expectation Maximization (EM) algorithm, and implements a two-layer decision through the utility function of the ID model. Reference [27] proposes the Sequential Neural Variational Agent (SeNeVA) variational Bayesian mixture model, which models the future trajectory distribution as a multivariate Gaussian mixture, estimates the mixture coefficients using an allocation network, generates representative trajectories through non maximum suppression sampling, and quantifies motion prediction uncertainty.

The generalization ability of Bayesian Neural Networks (BNNs) in real - world scenarios has also been systematically verified. Reference [90] generates calibration predictions and uncertainty estimates in three typical scenarios: urban navigation, obstacle avoidance, and weather visual degradation. Its probability output supports conservative interpretable decisions in high-risk environments; Reference [91] introduces Bayesian uncertainty quantification method in the field of deep learning, which distinguishes safe and failed driving behaviors by calculating the uncertainty score of vehicle driving process in system level simulation testing. The Pyramid Bayesian deep learning method in the reference [92] simplifies and reduces the Monte Carlo dropout layer and introduces a pyramid pooling module to quantify the uncertainty of semantic segmentation models on the Cityscapes dataset. The noise-aware decision-making (NADM) algorithm proposed in the reference [93] integrates Bayesian neural networks and oblique geometry Jensen Shanson divergence to construct a new loss function optimization network, quantifying the uncertainty caused by system noise and function approximation errors.

Furthermore, Bayesian methods have also been applied in structural risk modeling. The scenario based risk method based on Bayesian networks in the reference [94], combined with safety performance indicators (SPIs) and uncertainty estimation, quantifies residual risk through conditional probability tables and preserves the dependency relationships of vehicle elements. Reference [95] integrates Fault Tree Analysis (FTA) and Bayesian Networks (BN) to construct a fault tree with collision as the top event. Bayesian inference is used to update the posterior probability and quantify the impact of subsystem failures on collision risk.

4.4.2 Sequence model modeling method

In the research on risk quantification for autonomous driving, deep neural network methods based on sequence modeling have been widely applied to tasks such as driving behavior analysis, driver state recognition, and environmental risk prediction, emerging as an important technical means to support system - level safety decision - making. By modeling temporal dynamics, these methods can effectively capture the interaction relationships and potential uncertainties among multiple stages and multiple participants.

One category of studies focuses on the temporal safety assessment mechanism of the system itself. For example, the reference [96] proposes the concept of self-evaluation for autonomous driving, quantifying risks by comparing three types of time constraints: emergency braking time calculated by physical constraints, driving maneuver duration constraints, and reliable prediction horizon of deep learning trajectory prediction models. The operating domain is divided into comfort, safety, and unsafe levels to achieve safe driving. Reference [97] research based on dynamic Bayesian networks (DBNs) utilizes vehicle multi-sensor time-series data to learn a model of discrete node variables, and uses Markov jump particle filtering (MJPF) tool to predict future states based on the learned DBNs. The risk is quantified by detecting potential abnormal behavior of vehicles.

Another category of studies focuses on risk identification in interactive behaviors and driver state perception. Reference [98] uses Long Short Term Memory (LSTM) neural networks to recognize drivers' lane changing intentions, combined with Lightweight Gradient Boosting Machine (LGBM) algorithm to predict lane changing risks. Key influencing factors are identified through feature importance analysis, and lane changing scenario risks are quantified to support safety decisions. In terms of driver state monitoring, the reference [99] proposes a real-time driver drowsiness detection framework, based on the Transformer architecture, using models such as Vision Transformer (ViT) and Swin Transformer to classify eye states, combined with Haar cascade classifiers to extract facial and eye regions, and improving model interpretability through Class Activation Mapping (CAM) to quantify driver state risks and trigger alerts in a timely manner to ensure safety.

As uncertainty modeling is gradually introduced into neural network architectures, studies have further attempted to combine Bayesian estimation mechanisms with Transformer structures. Reference [100] proposes the Uncertainty based Transformer (UBT) model, which

quantifies the feature uncertainty of the backbone network through a dual Gaussian feature map network (DGF), screens high-quality query vectors using an RBF based query filtering model (RBQF), and outputs the quantified uncertainty through an uncertainty detection head (UDH). In addition, the reference [101] decomposes the scenario into different participant activities and environmental conditions, estimates the severity of risks through Monte Carlo simulation, calculates exposure probability based on real driving data, and quantifies driving scenario risks to support safety measures.

For uncertainty control at the strategy layer, current studies tend to integrate multi - source information modeling and deep fusion mechanisms into a unified framework to enhance the risk response capability of the system. Reference [102] proposes the Risk Aware Decision Making (RADM) framework, which quantifies cognitive uncertainty and estimates the risk of prediction failure through a multi-agent prediction network integrating traffic light features and deep integration techniques. Based on the evidence-based deep learning paradigm combined with Beta priors, the reference [103] improves the interpretable target induction model by integrating uncertainty evaluation into training strategies guided by data reweighting, enhancement, and other uncertainties, thereby enhancing decision transparency and reliability. Reference [104] proposes a driving risk assessment framework that integrates multimodal data, uses the latent Dirichlet allocation (LDA) model to classify risk levels based on driving behavior characteristics, and integrates "low-level" historical sequences and "high-level" statistical features of multiple features through a spatiotemporal dual channel neural network (MLA-DCNN) based on a multi-layer attention mechanism to capture the differences in spatiotemporal and extracted dimensions of features. Reference [105] analyzes the effects of non driving related tasks (NDRTs) with different resource requirements and participation durations on driver physiological responses (pupil diameter, heart rate) and takeover performance through driving simulator experiments, quantifying the risks caused by driver state changes.

Reference [106] proposes that the ThirdEye tool generates attention maps through interpretable artificial intelligence technology and converts them into confidence scores, combined with probability distribution fitting to determine threshold quantification risk. The unified prediction and planning framework of the reference [107] utilizes Gaussian mixture models and deep integration methods to simultaneously model short-term random uncertainty, long-term random uncertainty, and cognitive uncertainty. Through a specially designed risk model, uncertainty aware planning is achieved to improve planning accuracy and reliability. Reference [108] is a driving risk assessment framework based on entropy aware graph neural network (EIGNN) and Gaussian distribution (GD). GD is used to probabilistically model vehicle speed and acceleration, and entropy theory is introduced to quantify risk uncertainty. Graph neural networks are used to capture the spatiotemporal dynamics of multi vehicle interactions and predict the potential risk level of driving strategies.

In the research of risk quantification in autonomous driving, the application of diversified learning paradigms provides multiple technical paths for risk assessment and safety decision-making. In the field of imitation learning, the reference [109] estimates the output uncertainty of end-to-end control systems through Monte Carlo dropout method, and selectively obtains new training data under uncertain states to quantify the risks in autonomous driving; Meanwhile, this method actively avoids potential error states, providing support for achieving safe autonomous driving. In terms of contrastive learning, the Drive CLIP framework proposed in the reference [110] innovatively integrates text descriptions and event data through cross modal contrastive learning to construct a representation system for safety critical driving scenarios; By utilizing natural language supervision to extract scene event embeddings, event data can support text guided zero sample driving risk analysis, expanding the application dimensions of risk quantification in sceario understanding.

4.4.3 Reinforcement learning method

Facing the risk challenges brought by high dynamics and high uncertainty in open road environments, traditional methods based on rules or supervised learning are difficult to cover all risk patterns in complex scenarios. In this context, reinforcement learning has shown unique advantages: its online learning and strategy adjustment mechanisms enable it to autonomously optimize decision - making strategies in constantly changing environments. Meanwhile, by introducing uncertainty estimation and risk index evaluation, it can realize the identification and intervention of high - risk scenarios, providing a new technical pathway for safety modeling.

Reference [111] proposes the intervention risk classification twin delayed deep deterministic policy gradient-task decompose (IDRCTD3-TD) algorithm, which decomposes driving tasks by constructing road and vehicle risk potential fields, combines low-risk episode sampling enhancement methods to improve sample utilization, and adopts an intervention training strategy that incorporates artificial potential fields to quantify autonomous driving risks. In addition, the Ensemble Quantile Networks (EQN) method proposed in the reference [15] combines distributed reinforcement learning with ensemble methods. It quantifies random uncertainty by learning the quantile function of the reward distribution and trains the agent ensemble based on bootstrap data to achieve Bayesian estimation of model uncertainty. It also introduces uncer-

tainty classification criteria to provide quantitative basis for safety decision-making.

Reference [112] proposes a method based on supervised ensemble deep deterministic policy gradient (SE-DDPG) to construct a continuous action space reinforcement learning strategy. Multiple comment networks are integrated to quantify model uncertainty, and a switching mechanism is designed to activate the principle strategy based on intelligent driver model (IDM) and lane change minimum total braking (MOBIL) when the model uncertainty exceeds the safety threshold, ensuring decision safety. The Uncertainty Aware Model Based Reinforcement Learning (UA-MBRL) method proposed in the reference [113] evaluates the uncertainty of the environmental model by constructing an action condition ensemble model, dynamically adjusts the prediction horizon based on an adaptive truncation mechanism, and achieves dynamic adaptation of risk quantification. Reference [28] proposes a risk-sensitive reinforcement learning framework named RACER, which constructs an integrated distributed comment network to quantify both aleatoric and epistemic uncertainties. The actor network is optimized using Conditional Value at Risk (CVaR) as the objective function, while an adaptive action space constraint mechanism dynamically adjusts behavioral boundaries. This framework enhances the precision of risk estimation and ensures the safety of decision-making in autonomous driving scenarios.

5 Perspective on data source and assessment framework

5.1 Different data sources

The source and quality of data constitute the foundation for the effectiveness and scalability of risk quantification methods, directly determining the accuracy, reliability, and applicable scope of assessment results. Classifying risk quantification studies from the perspective of data sources helps to understand the capability boundaries and applicable scopes of different assessment strategies. In this section, data are divided into three main source pathways:

Firstly, the risk scenario screening method based on large - scale driving data excavates extreme or typical high - risk situations from historical data through statistical modeling and failure mode detection. It is suitable for building a risk database with wide coverage and practical reference value. Secondly, the data generation method based on simulation synthesizes rare but critical long - tailed high - risk situations through high - fidelity simulators. This effectively makes up for the difficulty in collecting high - risk data in the real world and provides more diverse training and verification samples for algorithms. Finally, the risk labeling method based on human observation mainly relies on multimodal sensors (such as eye trackers, electroencephalographs, facial recognition cameras, etc.) to collect driver state and behavior information. It achieves risk expression that is closer to human cognitive mechanisms by inferring drivers' attention distribution, behavioral intentions, and reaction durations.

Regarding research on screening high - risk scenarios from large-scale real-world driving data, Current research focuses on efficiently identifying safety-critical scenarios from massive real-world driving data. Reference [29] introduces a real-world road corner case dataset for object detection in autonomous driving that automates the detection of corner cases by analyzing physical occlusion likelihood (via LiDAR clustering) and perceptual novelty (using failure modes of pretrained detectors), reducing manual curation by 90% and compiling the diverse dataset. Complementarily, Reference [30] develops a probabilistic risk model to filter high-value scenarios from datasets like Waymo Open Motion, quantifying collision risks by modeling agent trajectory uncertainties as type-dependent Gaussians, computing spatiotemporal overlap probabilities, and integrating time-dependent risks via survival functions with hierarchical classification.

Regarding the active construction of risk scenarios, Current research emphasizes actively generating high-risk driving scenarios via simulation to address data scarcity. Reference [32] leverages Causal Bayesian Networks (CBN) derived from real accident data to identify high-risk static patterns and generate hazardous scenarios in CARLA, significantly reducing iteration cycles. Meanwhile, Reference [31] proposes the DeepMF framework, which decomposes collision risk using Bayesian probability for closed-loop risk maximization, enabling realistic adversarial maneuvers. Furthermore, Reference [114] employs Genetic Algorithms, defining Scenario Potential Risk (SPR) as a fitness function to evolve road structures and trajectories, efficiently synthesizing complex disturbances.

Regarding research on risk labeling based on human feedback, increasing attention has been paid in recent years to associating structured subjective evaluations with objective risk indicators. Reference [115] quantifies driver states in conditional automation by collecting standardized scores (e.g., KSS for drowsiness, DoT for takeover difficulty), demonstrating significant correlations with objective risks like loss-of-control rates. Similarly, Reference [116] evaluates human-AI risk in Tesla's ADS by measuring subjective uncertainty tolerance through observed interventions and trust expressions, linking tolerance variations to differential risk outcomes. Reference [17] proposes an intelligent risk assessment system and an AI driven risk information human-machine interface, integrating machine learning, deep learning, and large language models to monitor users' neurocognitive, neuropsychological, and neurophysiological states in real-time. Combining the state of the auto drive system and environmental factors, the collision risk is quantified into three levels: low, medium and high, and users are guided to take responsible actions through emotional interaction to form a risk

quantification scheme that takes into account both technical safety and user perception.

5.2 Risk assessment framework

After identifying risk sources, selecting quantification methods, and establishing corresponding data support, a further key issue is how to effectively integrate the aforementioned elements into the system architecture to form a verifiable and deployable closed-loop risk management mechanism. Classifying risk quantification studies from the perspective of system architecture helps to clarify the logical relationships, data dependencies, and responsibility divisions among various functional modules, thereby constructing a complete safety assessment and control system. This section mainly focuses on the following types of research pathways: The evaluation system based on predefined indicators quantifies the risk level by setting physical or statistical rules; The risk boundary identification method based on the Operational Design Domain (ODD) explores the capability scope and safety constraints of the system under actual deployment conditions; The functional safety framework based on SOTIF (Safety of the Intended Functionality) conducts modeling analysis and reasoning decomposition for insufficient functionality and unknown risks.

Firstly, in the aspect of risk indicator construction, relevant studies are dedicated to establishing a quantifiable standard system from multiple dimensions to support the systematicness and operability of risk assessment. Reference [117] introduces the PODAR model, a physics-based framework quantifying collision severity through interpretable parameters, calibrated via obstacle-avoidance data to project damage risk. Simultaneously, Reference [118] develops a real-time Bayesian framework integrating TTC to compute probabilistic crash risk indices and introduces a novel Risk Exposure (RE) metric. Complementing probabilistic approaches, Reference [119] proposes an evaluation framework for safety metrics, using collision unavoidable moments derived from trajectory data to objectively benchmark real-time metrics via PR curves. Reference [120] defines risk as the product of collision probability and expected harm, establishing a human-derived threshold RmaxRmax to invalidate unsafe trajectories during planning. In parallel, Reference [121] conceptualizes risk as the control effort required for collision avoidance, mirroring human responses to achieve unified perception-action quantification. Operationalizing runtime safety, Reference [122] aggregates sensor-level risks (health/data quality) into a unified metric via SVM-learned weights, triggering fail-safe strategies when thresholds are exceeded. Finally, to validate all approaches, Reference [123] creates a trajectory-based framework defining risk through optimization-derived "collision unavoidable" events, providing objective ground truth for metric calibration.

For scenario complexity, Reference [124] links crash severity to environmental features via SMOTE/XGBoost, while Reference [125] fuses static entropy (Bayesian networks) and dynamic interactions (gravitational models) into a tiered framework, later refined by Reference [36]'s DSCQ method multiplicatively combining static/dynamic uncertainty. Simultaneously, Reference [37] employs multi-agent reinforcement learning (MARL) with Social Value Orientation to quantify ramp conflict resolution difficulty. Finally, Reference [35] quantifies drivability through a composite framework integrating decision confidence (tactical drivability), maneuver adherence (operational drivability), and trajectory quality metrics (motion-quality drivability), using a unified scoring function to balance strategic intent, motion feasibility, and ride comfort.

To ensure cross-environment reliability, Reference [126] addresses domain shift challenges with DGCCL—a CLIP-based visual encoder optimized via center loss—enhancing detection robustness for consistent risk assessment under varying conditions. Expanding beyond distraction, Reference [127] employs XGBoost and SHAP to interpretably quantify situational awareness (SA) deficits during non-driving tasks, using interval-based information granules to capture SA uncertainty and identifying critical risk factors (e.g., gaze dispersion) via feature importance analysis. Finally, operationalizing insights, Reference [33] introduces a nonparametric DEA framework correlating dangerous behavior frequency with driving distance, deriving a dynamic "safety efficiency" metric for fleet-wide driver risk ranking and targeted interventions.

In recent years, research on the Operational Design Domain (ODD) of autonomous driving systems has received increasing attention, with the focus on evaluating the overall risk boundaries of the system, especially the identification of capability boundaries driven by uncertainty. Reference [38] quantifies perception ODD limits by integrating out-of-distribution (OOD) detection (energy scores, semantic-aware techniques) to flag safety-critical unknowns. Simultaneously, Reference [39] proposes a virtual validation framework using Monte Carlo sampling to model aleatoric/epistemic uncertainties (e.g., pose/heading errors), evaluating robustness via pass rates and trajectory errors to derive ODD thresholds like minimum safe overtaking distances.

Beyond the Operational Design Domain (ODD), research on the Safety of the Intended Functionality (SOTIF) further focuses on the potential risks caused by insufficient functionality or environmental errors in the system under fault - free conditions. Different from traditional functional safety that focuses on hardware or software faults, SOTIF emphasizes the modeling and assessment of risks caused by "unknown unknowns" and functional limitations, serving as an important supplement to the safety assurance system of autonomous driving systems.

Therefore, how to systematically identify and quantify the deficiencies of intended functionality and construct risk models oriented to SOTIF is the current research focus. Reference [40] propagates perception uncertainty via Deep Ensembles and SOTIF entropy to dynamically adjust MPC risk fields, while Reference [46]'s EviSeg leverages evidence theory to flag segmentation errors via conflict metrics. Alternatively, Reference [128] applies noisy hierarchical Bayesian networks to quantify unobserved triggers using real-time IoV data. For system-level decomposition, Reference [129] allocates safety constraints to sensors via Shapley value-based game theory. Concurrently, Reference [130] develops multi-field risk models (potential/kinetic/behavioral) and ontological complexity metrics. Reference [41] formalizes SOTIF risk probabilistically while Reference [131] refines ISO 21448 via causal chains (trigger→harm) bounded by Bonferroni inequality. Further advancing validation, Reference [42] integrates quantitative FTA with Safety Performance Indicators, using Monte Carlo sampling to compute residual risk.

6 Challenges and potential solutions

In the previous sections, we have discussed autonomous driving risk assessment methods from three different perspectives: risk sources, risk assessment methods, data sources, and risk assessment frameworks. We have noted that due to the highly complex dynamic environments involved in autonomous driving tasks, achieving safe and reasonable risk assessment remains a difficult problem. In this section, we further summarize three core challenges in current research and conduct in-depth analyses on each issue to reveal the limitations of existing methods and provide theoretical insights and practical directions for future research.

Challenge 1: A unified standard for the risk assessment indicator system has not yet been formed. Currently, common practices involve using some physical or statistical indicators (such as Time to Collision (TTC), collision energy, acceleration, etc.) to measure risk, but these indicators often struggle to fully reflect the real safety level in driving tasks. For example, in some scenarios, although the TTC is very short, the movement trends between vehicles will not actually lead to a collision; while in other scenarios, although they seem safe on the surface, they are prone to perception or behavioral misjudgments. In addition, some studies attempt to introduce drivers' subjective evaluations (such as fatigue level, willingness to take over, etc.) to assist in risk judgment, but such data are difficult to standardize and vary greatly among different individuals. Therefore, how to establish an assessment indicator system that can both reflect actual risks and have universality remains a key issue.

To address the challenges of risk assessment in autonomous driving, future research should focus on developing a comprehensive and dynamically adaptive risk assessment framework. This framework should integrate physical models, driver behavior models, and perception data, considering environmental information and contextual factors to achieve accurate assessment in complex driving scenarios. By incorporating multidimensional data sources (such as sensor information, driver intent, etc.), combining task-oriented evaluation methods (such as lane-keeping, lane change, etc.), and dynamically adjusting evaluation results based on real-time context (such as weather, lighting, traffic density), the accuracy and generalizability of the risk assessment can be significantly enhanced, thereby providing safer and more reliable decision-making support for autonomous driving systems.

Challenge 2: Risk depends not only on the environment but also on the capabilities of the autonomous driving system itself. In the same driving scenario, if the system has accurate perception, reasonable decision-making, and stable control, the risk it faces may be very low; however, if the system's capabilities are weak, dangers may occur even in seemingly simple environments. Nevertheless, most risk assessment methods tend to only focus on the danger level of the scenario itself and ignore differences in system capabilities. This will lead to distortion of assessment results, failing to accurately reflect the real risk levels of different systems. Therefore, assessment methods need to be able to dynamically adjust risk judgments based on the perception, decision-making, and control capabilities of the autonomous driving system.

To address this challenge, future assessment methods should be capable of dynamically adjusting risk judgments based on the capabilities of the autonomous driving system. By monitoring the system's perception, decision-making, and control capabilities in real time, the risk assessment results can be adjusted according to the current system performance. For example, when the system's perception or decision-making capabilities are weak, the risk assessment should automatically increase to reflect potential dangers. Additionally, by incorporating adaptive learning algorithms, the system can continuously optimize the risk assessment model during actual driving, thereby improving the adaptability and accuracy for different system capabilities.

Challenge 3: Data used for risk assessment is difficult to obtain, especially data with accurate risk labels. One way is to screen out pre-accident dangerous scenarios from real road data, but the quantity of such data is extremely small, and evaluation criteria often need to be artificially set, such as "relative speed exceeding a certain threshold and distance being too close", etc., and these rules have limitations. Another way is to indirectly judge the risk state by monitoring driver behaviors (such as eye movement trajectory, heart rate, intervention actions), but this method relies on a large number of sensors, with high costs and difficulties in data synchronization, making it difficult to ap-

ply on a large scale. Therefore, how to efficiently and low-costly construct large-scale datasets with risk labels is one of the main obstacles currently restricting risk modeling and algorithm training.

To address this challenge, future research should explore methods to efficiently obtain risk-labeled data, such as generating data through simulated environments, collecting real-time data with onboard sensors, and utilizing crowdsourced data collection. By combining simulated scenarios with real-world data, risk-labeled datasets can be generated on a large scale. Additionally, using onboard sensors to monitor driver behavior and vehicle status in real driving conditions allows for dynamic labeling of data. Crowdsourcing driving data from vehicle owners can also help expand the dataset while reducing costs. These approaches will effectively overcome data limitations and advance risk modeling and algorithm training.

7 Conclusion

This review comprehensively examines the development context and latest progress of research related to risk assessment in the field of autonomous driving. From three core dimensions—risk sources, modeling methods, and data & assessment frameworks—the article conducts systematic classification and in-depth analysis of existing works, and explores the technical routes and applicable boundaries of current mainstream methods. On this basis, it further summarizes three key challenges restricting the construction of risk assessment systems: first, a unified standard for the risk assessment indicator system has not yet been formed, making it difficult to fully reflect the actual risk level in the driving process. Second, existing methods fail to fully consider the impact of capability differences of autonomous driving systems on risk levels, making it hard to achieve dynamic assessment based on system capability differences. Third, the acquisition of high-quality labeled risk data is highly challenging, which seriously restricts the training breadth and verification depth of models. It is hoped that this review can provide systematic theoretical support and methodological guidance for constructing an autonomous driving risk assessment framework with high reliability, strong generalization, and scalability.

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Author contributions
Dongyuan Lu conducted part of the literature survey and contributed to the writing of the initial draft. Haoyang Du and Zhengfei Wu assisted in organizing and managing the literature materials. Shuo Yang conceived the survey framework, led the literature review and analysis, wrote the majority of the manuscript, and finalized the submission. All authors read and approved the final manuscript.

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Data availability
The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests
The authors declare that they have no competing interests.

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